IDENTIFICATION, SEMANTIC COARSENING, AND CERTIFIED INVERSION IN SPARSE MULTI-VIEW LATENT CLASS MODELS

Dedicated to those who fight harder when everything is turning against them.

Abstract. Sparse multi-view data can identify latent mixture components while leaving their scientific interpretation unresolved. We study finite categorical mixtures observed through a covered family of triples, with shared group kernels and componentwise mixture weights. Under local uniqueness, we describe the entire observational fiber by compatible edge permutations and weight-sensitive overlap stabilizers. Kruskal's condition reduces this fiber to independent permutation orbits on the connected components. A fixed binary coarsening of the latent states then yields a separate semantic identification problem. We give an exact surviving-permutation criterion and a full-rank family in which one strict direction anchor fails even when the entire joint output law is known. For positive diagonal observations, we characterize the unrestricted fixed-state-count real fiber and obtain attained, sharp sensitivity, specificity, and prevalence images by finite subset enumeration. On a uniformly conditioned parameter class, explicit Jacobian and curvature bounds give local inversion. For algebraically specified margins, a terminating rational interval-cover algorithm computes a positive global separation bound with an independently checkable certificate. An explicit leaf-count lower bound shows that this certificate format can be prohibitively large. Any valid separation bound yields global inverse bounds and finite-sample subclass rates, although the stated sufficient sample rule can be highly conservative. Gold-direction certification and confidence sets under bounded conditional-dependence error complete the statistical interpretation. The core identification, conditioning, statistical, and certificate-correctness results are accompanied by Lean formalizations. Computability is established; practical full-model certification is not demonstrated.

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1. Introduction

A set of readers, sensors, or predictive systems may be evaluated on overlapping subsets of cases rather than on one common complete sample. If several outputs share a source of dependence, it is natural to combine them into a single categorical group output. Three such groups provide a tensor of joint probabilities. Across a sparse assignment design, the same group may appear in several tensors, making agreement of its latent columns a source of information about how local decompositions fit together.

There are two identification questions in this setting. The first asks which latent product components and weights are determined by the observations. The second asks which components correspond to a scientifically defined class, such as a binary target condition. The second question remains meaningful after the first has been answered. In particular, a product decomposition conditional on a latent subclass Z need not be a product decomposition conditional on a coarser truth variable T. Averaging subclasses within a truth class can induce dependence between groups.

This paper develops these two questions in one fixed finite-state model. The design is a nonempty collection of observed triples covering all included groups. Each group has a stable categorical kernel, and each connected component of the edge-intersection graph has its own positive mixture weights. The number r of positive mixture terms is fixed in both the reference model and its competitors. Parameters are real, including at stochastic-kernel boundaries.

The results form the following chain.

(1) Given local uniqueness, Theorem 3.2 identifies the full fiber with an assembly of compatible edge permutations. Compatibility uses both the weights and the kernels on shared groups. It yields an exact criterion for identification modulo any declared subgroup of the componentwise permutation group and any fixed information predicate.
(2) Under Kruskal's condition, Theorem 3.4 specializes this description to one free permutation per connected component. Theorem 4.1 determines when external information reduces these permutations to an appropriate semantic subgroup. Theorem 4.2 shows why a direction anchor need not do so when a truth class contains several latent subclasses.
(3) Theorem 4.3 characterizes the full real fixed-r fiber of a positive diagonal observation family. The resulting sharp semantic image in Theorem 4.4 is a finite set with attained extrema. In a four-state example, its exact sensitivity image is $\{1/2, 1\}$.
(4) Theorem 5.2 supplies explicit normalized-coordinate Jacobian and curvature constants from lower weight and singular-value margins. Theorem 6.4 computes a positive separation bound by a rational cover search whose certificates are valid for every real admissible pair. Any valid separation bound then turns local inversion into the global bounds of Corollary 5.4. Proposition 6.5 also identifies a severe output-size limitation of the explicit box certificate.
(5) Theorem 7.1 composes these deterministic results with finite-block sampling and a stated near-global fitting guarantee. Propositions 7.6 and 7.7 connect semantic direction information and approximate conditional independence to confidence statements.

The statistical loss in the inversion results is explicitly a loss for unnamed subclasses. Semantic performance is assessed through its own fiber or confidence-set image.

1.1. Relation to earlier work.

Latent-class analysis of repeated ratings has a long history. Dawid and Skene [5] estimate observer error rates in a latent-class framework. Three-way tensor uniqueness is governed by the classical theorem of Kruskal [13]; Allman, Matias, and Rhodes [1] explain its normalized latent-structure consequences. We use this theorem as a

standard local ingredient; its stochastic form is the normalization in [1, Corollary 2]. The accompanying general Kruskal proof follows the rank, permutation, and alignment argument of Stegeman and Sidiropoulos [22]. The explicit full-column-rank proof in Appendix A also makes the quantitative branch independent of the general Kruskal argument.

Coupled tensor uniqueness is developed by Sørensen and De Lathauwer [21]. Recovering a joint probability model from lower-order marginals and coupling tensor factors across those marginals are established approaches. Kargas, Sidiropoulos, and Fu [12, Theorems 1–2] give generic joint-distribution recovery conditions from low-order marginals. Flores, Usevich, and Brie [6, 7] develop coupled tensor models for probability mass function estimation, including subsets of triple marginals, hypergraph coupling, uniqueness, and Jacobian-rank questions. Our local-to-global statement records the exact full fiber after local rigidity is established, including weight-sensitive overlap stabilizers, a specified residual gauge, and arbitrary fixed external information. The assumptions are complementary: coupled uniqueness can hold without a unique decomposition on every individual triple. For example, [7, Theorem 8.6] transfers identifiability of a reduced real CP model to a Cartesian coupling, whereas Theorem 3.2 here assumes local rigidity and determines the exact parameter-specific residual fiber. This interface is used subsequently for scientific coarsening and certification. Permutation synchronization also appears in multi-way matching [18]; the overlap relations here are exact stabilizer constraints derived from the mixture parameters.

Graph-based identifiability and learning guarantees also arise in crowdsourcing. Ma, Olshevsky, Szepesvári, and Saligrama [14] analyze sparse rank-one worker-interaction models. Zhang, Chen, Zhou, and Jordan [26] combine three-group spectral initialization and expectation maximization with finite-sample guarantees. Their assumptions and losses differ from the grouped categorical model considered here. In particular, the event-direction information used below is weaker than a complete assignment of all product components to named scientific classes.

Subclasses nested within scientific classes are a substantive modeling device for dependence, as in Wu, Deloria-Knoll, and Zeger [24]. Jaffe et al. [11] also introduce an additional latent layer to model dependent classifiers; their dependence structure is related but is not the same coarsening model. Hui and Walter [10] identify diagnostic-test errors using population variation and conditional-independence assumptions. Pepe and Janes [19] discuss the interpretation of diagnostic latent classes. Partial identifiability under restricted response models is studied by Gu and Xu [8], and sharp diagnostic-performance sets with imperfect reference tests by Obradović [17]. Here the semantic ambiguity can persist even when the product components themselves have distinct, perfectly conditioned kernels. The diagonal family provides an explicit full-fiber calculation of that ambiguity under a fixed truth-block profile and strict event anchors. Its tensor uniqueness is a classical special case; the semantic image is the object of the calculation.

Tensor conditioning through singular values of derivative maps is studied by Vannieuwenhoven [23]. Quantitative stability of tensor decompositions under robust Kruskal conditions is established by Bhaskara, Charikar, and Vijayaraghavan [4]. Bhargava, Schulman, and Sivan [3] give a randomized exact-real-RAM decomposition algorithm under Kruskal's condition. Our contribution to this part of the theory is an explicit set of constants in full normalized coordinates, its covered-design extension, and a verified rational procedure for a positive global separation bound. The program constructs a positive certificate rather than an exact minimum or a fitted decomposition; its guaranteed termination is not a practical resource bound. Its interval-cover principles belong to established verified computation [16, 20]. Likewise, the

probability bounds use Hoeffding's inequality [9]; the specific content is their composition with the stated model, gauge, fitting condition, and semantic information.

The resulting contribution is a unified treatment of exact residual fibers, semantic partial identification, explicit conditioning, and machine-checked positive-separation computation.

2. The model and its gauges

2.1. **A covered triple design.** Let G be a finite set of groups and let $\mathcal{E}$ be a nonempty set of three-element subsets of G such that $\bigcup_{e\in\mathcal{E}} e = G$. A group g has a finite nonempty output alphabet Y_g, of size n_g. This alphabet may encode a joint pattern of several dependent outputs within the group. Set $X_e = \prod_{g\in e} Y_g$ and $d_e = |X_e|$.

The edge-intersection graph has vertex set $\mathcal{E}$, with distinct edges adjacent when they share a group. Write $\mathcal{C}$ for its set of connected components and $[e]$ for the component of e. All edges incident to a given group are adjacent or equal, so they belong to one component. This defines $c(g) \in \mathcal{C}$ without a choice of incident edge. The model has no observed cross-component joint law or imposed equality of weights between distinct components.

Fix $r \geq 2$ and $Z = \{1, \dots, r\}$. A parameter $\theta \in \Xi_r$ consists of real arrays

$$w_c(z) > 0, \quad \sum_z w_c(z) = 1, \qquad L_g(y,z) \geq 0, \quad \sum_{y\in Y_g} L_g(y,z) = 1. \tag{2.1}$$

There is one weight vector per component and one kernel per group. The observation map is

$$\mathcal{O}(\theta)_e(x) = \sum_{z=1}^{r} w_{[e]}(z) \prod_{g\in e} L_g(x_g, z), \qquad e \in \mathcal{E},\ x \in X_e. \tag{2.2}$$

Thus the three group outputs on an edge are conditionally independent given Z. Shared arrays in (2.2) express the required stability across incident triples. The integer r is the number of strictly positive mixture terms. It is not, by definition, a minimal tensor rank. Every fiber in the paper compares parameters with this same r unless a different state count is explicitly stated.

2.2. **Componentwise permutations and observational fibers.** Let

$$W = \prod_{c\in\mathcal{C}} \mathrm{Sym}(Z).$$

For $q \in W$, define the simultaneous pullback

$$(S_q w)_c(z) = w_c(q_c(z)), \qquad (S_q L)_g(y,z) = L_g(y, q_{c(g)}(z)). \tag{2.3}$$

We use ordinary function composition, so $S_p(S_q\theta) = S_{q\circ p}\theta$. Pullback preserves (2.1) and (2.2). A declared gauge Γ is a subgroup of W; it determines which relabelings are considered equivalent for the question at hand.

Let $\mathcal{A}$ be a fixed predicate on Ξ_r, representing external information. It need not be invariant under W or under Γ. Define

$$\mathcal{F}(P, \mathcal{A}) = \{\eta \in \Xi_r : \mathcal{O}(\eta) = P,\ \mathcal{A}(\eta)\}.$$

For $\mathcal{A}(\theta)$, identification modulo Γ means

$$\mathcal{F}(\mathcal{O}(\theta), \mathcal{A}) = \{S_\gamma\theta : \gamma \in \Gamma\} \cap \{\eta : \mathcal{A}(\eta)\}. \tag{2.4}$$

The intersection retains the fixed information predicate after relabeling.

2.3. **Scientific coarsening.** Fix a surjection $\tau : Z \to \{0,1\}$ and write $m_t = |\tau^{-1}(t)|$. The interpretation of $T = \tau(Z)$ is external to the unlabeled decomposition. Define

$$\pi_c^T(t) = \sum_{\tau(z)=t} w_c(z), \tag{2.5}$$

$$K_g^T(y,t) = \frac{\sum_{\tau(z)=t} w_{c(g)}(z) L_g(y,z)}{\pi_{c(g)}^T(t)}. \tag{2.6}$$

Both denominators are positive. These are normalized truth-conditional marginal kernels; they do not imply conditional independence across groups given T.

For a readout event $R \subseteq Y_g$, sensitivity and specificity are

$$f_{g,R}(\theta) = \sum_{y \in R} K_g^T(y,1), \qquad s_{g,R}(\theta) = \sum_{y \notin R} K_g^T(y,0). \tag{2.7}$$

Truth prevalence is $\pi_c^T(1)$. For any named probability-valued functional f, its identified image is $I_f(P,\mathcal{A}) = \{f(\eta) : \eta \in \mathcal{F}(P,\mathcal{A})\}$. When this set is nonempty, its sharp interval hull is $[\inf I_f, \sup I_f]$, and its width is $\sup I_f - \inf I_f$. The image itself need not be an interval.

The truth-preserving gauge is

$$\Gamma_T^0 = \{q \in W : \tau(q_c(z)) = \tau(z) \text{ for all } c, z\}.$$

The larger semantic gauge Γ_T permits, in each component, either preservation or exchange of the two entire truth blocks:

$$\Gamma_T = \{q \in W : \forall c\ \exists a_c \in \{0,1\}\ \forall z,\ \tau(q_c(z)) = \tau(z) \oplus a_c\}.$$

Both are subgroups. If $m_0 \neq m_1$, a whole-block exchange cannot be a bijection, and $\Gamma_T = \Gamma_T^0$. Truth-preserving permutations leave all functionals in (2.7) and (2.5) unchanged. A whole-block exchange changes the named truth labels and need not preserve named sensitivity.

An event-direction anchor is a fixed triple (g, B, s) with $B \subseteq Y_g$ and $s \in \{-1, 1\}$, imposing

$$D_{g,B,s}(\theta) = s\left(\sum_{y \in B} K_g^T(y,1) - \sum_{y \in B} K_g^T(y,0)\right) > 0. \tag{2.8}$$

Its group, event, sign, and semantic interpretation stay fixed in every comparison. The anchor provides a direction, rather than atom membership, a known prevalence, or an exact conditional probability. A family of such inequalities is one possible predicate $\mathcal{A}$.

For the binary decision that reports positive on B, the contrast is

$$D_{g,B,s} = s\big(f_{g,B} + s_{g,B} - 1\big).$$

It is therefore a signed version of the usual Youden index [25]. Specifying its sign orients a binary decision, but need not determine which of several latent subclasses belong to each truth class.

3. Exact local to global identification

3.1. **Local rigidity and overlap stabilizers.**

Definition 3.1 (Local rigidity)**.** For a fixed $\theta \in \Xi_r$ and edge e, local rigidity means that every positive-weight, normalized r-term product representation of $\mathcal{O}(\theta)_e$ is obtained from the local weights and three kernels of θ by one common permutation of Z. The competing kernels are arbitrary real stochastic kernels; their ranks are not assumed.

Local rigidity is an explicit premise in the next theorem. It separates a local tensor question from the consistency constraints imposed by the design. For an edge and a pair of adjacent edges, respectively, define

$$H_e(\theta) = \{\sigma \in \mathrm{Sym}(Z) : w_{[e]} \circ \sigma = w_{[e]},\ L_g \circ \sigma = L_g\ (g \in e)\},$$
$$H_{ef}(\theta) = \{\sigma \in \mathrm{Sym}(Z) : w_{[e]} \circ \sigma = w_{[e]},\ L_g \circ \sigma = L_g\ (g \in e \cap f)\}.$$

Here $L_g \circ \sigma$ denotes column pullback. Both are finite subgroups. A tuple $q = (q_e)_{e\in\mathcal{E}}$ is compatible when

$$q_e \circ q_f^{-1} \in H_{ef}(\theta) \quad \text{for every adjacent pair } e, f. \tag{3.1}$$

Write $\mathrm{Comp}(\theta)$ for the compatible tuples.

Theorem 3.2 (Stabilizer description of the full fiber)**.** *Fix a covered triple design, $r \geq 2$, $\theta \in \Xi_r$, a fixed predicate $\mathcal{A}$ with $\mathcal{A}(\theta)$, and $\Gamma \leq W$. Suppose every observed edge is locally rigid at θ. Then:*

1. *Every $q \in \mathrm{Comp}(\theta)$ defines a unique parameter $\mathrm{Asm}_\theta(q)$ by using $w_{[e]} \circ q_e$ as the component weights and $L_g \circ q_e$ at any incident edge. This parameter has observation $\mathcal{O}(\theta)$.*
2. *With $\mathrm{Comp}_\mathcal{A}(\theta) = \{q \in \mathrm{Comp}(\theta) : \mathcal{A}(\mathrm{Asm}_\theta(q))\}$,*

$$\mathcal{F}(\mathcal{O}(\theta), \mathcal{A}) = \{\mathrm{Asm}_\theta(q) : q \in \mathrm{Comp}_\mathcal{A}(\theta)\}. \tag{3.2}$$

3. *Identification modulo Γ holds if and only if*

$$\forall q \in \mathrm{Comp}_\mathcal{A}(\theta)\ \exists \gamma \in \Gamma\ \forall e \in \mathcal{E}, \qquad q_e \circ \gamma_{[e]}^{-1} \in H_e(\theta). \tag{3.3}$$

Proof. For any array F indexed by Z and permutations a, b,

$$F \circ a = F \circ b \quad \Longleftrightarrow \quad F \circ (a \circ b^{-1}) = F. \tag{3.4}$$

Thus (3.1) is exactly equality of the pulled-back weights and overlap kernels. Weight agreement propagates along paths in the edge-intersection graph. All incident edges of one group are adjacent or equal, so their pulled-back kernels agree as well. This proves well-definedness of the assembled parameter. Each of its local systems is a common permutation of the original system, so reindexing the finite sum in (2.2) proves observation equality.

Conversely, let η belong to the full fiber. Local rigidity gives an edge permutation q_e matching its weights and three factors to those of θ. On adjacent edges, these local representations use the same weights and shared kernels of η. Equation (3.4) therefore gives compatibility. Coverage yields $\eta = \mathrm{Asm}_\theta(q)$, and the information predicate gives $q \in \mathrm{Comp}_\mathcal{A}(\theta)$. This proves (3.2).

Finally, $\mathrm{Asm}_\theta(q) = S_\gamma\theta$ if and only if all local weights and factors agree on each edge. By (3.4), this is exactly the last condition in (3.3). Every Γ-translate satisfying $\mathcal{A}$ already belongs to the fiber. The equivalence follows. □

The stabilizer tests involve the actual weights and overlap columns. They are finite mathematical equality tests, with effective evaluation requiring an appropriate representation of their entries. Outside the local-rigidity premise, additional local decompositions may occur and must be analyzed separately.

Example 3.3 (Weights can resolve an overlap stabilizer)**.** Take $r = 2$ and two edges $e = \{g, a, b\}$ and $f = \{g, c, d\}$. Let both columns of L_g equal $(1, 0)^\mathsf{T}$, and let L_a, L_b, L_c, L_d be I_2. With weights $(1/2, 1/2)$, each edge is locally rigid even though L_g has rank one. Indeed, the point-mass g marginal and positive weights force every g column to be $(1, 0)^\mathsf{T}$. The other two groups

have positive mass only at $(1,1)$ and $(2,2)$. Nonnegativity forces each product term to have singleton diagonal support; two positive terms covering two positive atoms give a bijection. Their weights are then forced by the two atom masses.

Let s be the nonidentity permutation of two states. The overlap stabilizer is $H_{ef} = \{\mathrm{id}, s\}$, whereas $H_e = H_f = \{\mathrm{id}\}$. The four pairs

$$(q_e, q_f) \in \{(\mathrm{id}, \mathrm{id}), (\mathrm{id}, s), (s, \mathrm{id}), (s, s)\}$$

are compatible and assemble to four distinct arrays. The common componentwise permutation gives only the two arrays indexed by $(\mathrm{id}, \mathrm{id})$ and (s, s). Thus the full fiber is strictly larger than $W \cdot \theta$.

If the weights are instead $(3/10, 7/10)$, the same support argument gives local rigidity, but a swap no longer fixes the weights. Hence $H_{ef} = \{\mathrm{id}\}$, only $q_e = q_f$ is compatible, and the fiber consists of the two common-permutation arrays. Figure 1 displays the equal-weight fiber and its quotient by W.

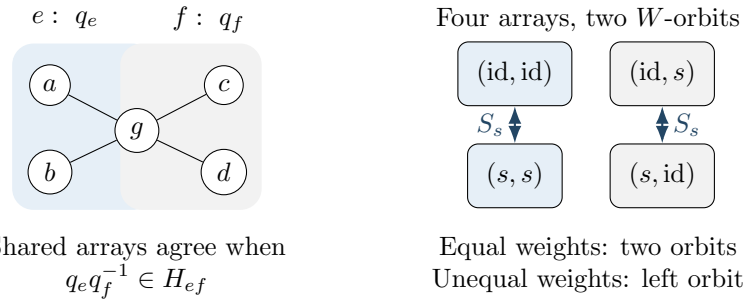


FIGURE 1. Local permutation choices are glued through the shared kernel and shared weights. In Example 3.3, the repeated overlap columns alone cannot synchronize the edges. Distinct weights remove the relative swap.

3.2. **The Kruskal specialization.** For a matrix with r columns, its Kruskal rank is the largest $k \in \{0, \ldots, r\}$ for which every k-column subfamily is linearly independent. Kruskal's three-factor theorem states that if the sum of the three reference Kruskal ranks is at least $2r + 2$, any other r-term decomposition agrees up to a common permutation and termwise scalings [13, 1]. Positive weights and stochastic normalization remove the scalings: absorb weights into one factor, normalize the other two, and recover the weights by column sums.

Theorem 3.4 (Componentwise subclass identification)**.** *Let $\theta \in \Xi_r$ satisfy*

$$\sum_{g \in e} \mathrm{krank}(L_g) \geq 2r + 2 \qquad (e \in \mathcal{E}). \tag{3.5}$$

Then all edges are locally rigid and

$$\mathcal{F}(\mathcal{O}(\theta), \mathrm{true}) = W \cdot \theta. \tag{3.6}$$

The action on this fiber is free. Consequently the fiber contains exactly $(r!)^{|\mathcal{C}|}$ distinct parameter arrays. Every competitor in (3.6) *ranges over the full Ξ_r.*

Proof. The normalized Kruskal theorem gives local rigidity. Each of the three ranks on an edge is at least two, because the other two are at most r. Coverage implies that every group has pairwise distinct columns. A permutation fixing one overlap kernel is therefore the identity,

so $H_{ef}(\theta) = \{\mathrm{id}\}$. Compatible edge permutations are constant on each connected component. Theorem 3.2 now gives (3.6).

If $S_q\theta = S_p\theta$, choose one group in each component. Its distinct columns give $q_c(z) = p_c(z)$ for every z. Hence $q = p$, proving freeness and the cardinality assertion. □

Full column rank at every group is sufficient for (3.5), since $3r \geq 2r + 2$. The Kruskal condition is broader: individual factors can be rank deficient while satisfying the rank-sum inequality. It remains a sufficient local condition, rather than a characterization of all identifiable tensors.

3.3. **Binary truth as a special case.** When $r = 2$ and each truth class contains one state, Z itself is the binary truth variable. Identify the two latent coordinates with $T = \{0, 1\}$ and define

$$\Theta = \{\theta \in \Xi_2 : \mathrm{rank}(K_g) = 2 \text{ for every } g \in G\},$$

where $K_g = L_g$ and $\pi_c = w_c$ are the kernels and weights with the two latent coordinates named by their truth values. The fixed design, strict weight positivity, and stochastic normalizations are inherited from Section 2. Product structure is then conditional on T. This is the setting in which a single valid direction can orient a component.

Corollary 3.5 (Exact residual binary swaps)**.** *Let every group kernel in Θ have rank two. Let $\mathcal{A}$ consist solely of a fixed family of component-local strict event-direction anchors, true at θ. If $C_{\mathcal{A}} \subseteq \mathcal{C}$ is the set of components touched by those records, then*

$$\mathcal{F}(\mathcal{O}(\theta), \mathcal{A}) = \{S_\varepsilon\theta : \varepsilon \in \{0,1\}^{\mathcal{C}},\ \varepsilon(c) = 0 \text{ for } c \in C_{\mathcal{A}}\}.$$

There are $2^{|\mathcal{C}\setminus C_{\mathcal{A}}|}$ distinct arrays. Labeled identification holds exactly when $C_{\mathcal{A}} = \mathcal{C}$. For a finite anchor family, at least $|\mathcal{C}|$ records are necessary, and one available valid record in every component is sufficient.

Proof. Theorem 3.4 gives exactly the componentwise binary swaps. Swapping a component negates every fixed anchor contrast supported there; its strict positivity excludes that swap. An uncovered component has no anchor constraint, so its swap survives. Freeness gives the count. A record touches only one component, yielding the structural lower bound. □

This count concerns semantic direction records in the stated information class. The number of sampled cases needed to certify a direction is a separate statistical quantity, addressed in Section 7.2.

4. Semantic ambiguity and sharp diagonal images

4.1. **The remaining permutation test.** Assume (3.5), fix τ, and let $\mathcal{A}(\theta)$ hold. Define the surviving permutations by

$$\mathrm{Perm}_{\mathcal{A}}(\theta) = \{q \in W : \mathcal{A}(S_q\theta)\}.$$

Theorem 4.1 (Semantic identification criterion)**.** *Under these assumptions,*

$$\mathcal{F}(\mathcal{O}(\theta), \mathcal{A}) = \{S_q\theta : q \in \mathrm{Perm}_{\mathcal{A}}(\theta)\}.$$

Identification modulo Γ_T holds if and only if $\mathrm{Perm}_{\mathcal{A}}(\theta) \subseteq \Gamma_T$. The corresponding truth-preserving statement holds with Γ_T^0. If the latter inclusion holds, every truth marginal and every named functional in (2.7) *agrees throughout the fiber.*

Proof. Intersect (3.6) with the fixed predicate. If all surviving permutations belong to the stated subgroup, the desired fiber equality follows. Conversely, identification supplies, for any surviving q, some γ in that subgroup with $S_q\theta = S_\gamma\theta$. Freeness in Theorem 3.4 gives $q = \gamma$. Truth-preserving pullbacks reindex each sum in (2.5)–(2.6), proving the final assertion. □

For a particular functional, constancy is exactly constancy of $q \mapsto f(S_q\theta)$ on the surviving set. It may hold even when the entire semantic parameter is unidentified. The criterion therefore keeps functional identification distinct from identification of all truth-labeled arrays.

Theorem 4.2 (A full-rank semantic obstruction)**.** *For every $h \geq 2$, every finite covered connected triple design, every balanced truth partition on $r = 2h$ states, and every anchor group g_a, take $Y_g = \{1, \dots, 2h\}$ for all g. There exist $\theta, \eta \in \Xi_{2h}$ such that:*

1. *all weights are $1/(2h)$, and every group kernel is a permutation matrix;*
2. *all edge laws agree, as do the full joint product-mixture laws on all groups;*
3. *both parameters satisfy the fixed anchor $(g_a, \{1\}, +1)$ with gap $1/h$;*
4. *for $R = \{1, 2\}$, their sensitivities are $2/h$ and $1/h$;*
5. *θ and η are not equivalent under Γ_T.*

Proof. Enumerate the truth-one coordinates as $t_1, \dots, t_h$ and the truth-zero coordinates as $f_1, \dots, f_h$. Define a bijection b by $b(t_i) = i$ and $b(f_i) = h + i$. Define b' by exchanging only the images of t_2 and f_1. Use equal weights and, at every group, the kernels $L_g(y, z) = \mathbf{1}\{y = b(z)\}$ and $L'_g(y, z) = \mathbf{1}\{y = b'(z)\}$.

Every observed edge has mass $1/(2h)$ at each all-equal cell (j, j, j) and zero elsewhere. The full joint law likewise has mass $1/(2h)$ at each constant output vector. Every kernel has all singular values equal to one. The truth-one atom sets are

$$S = \{1, \dots, h\}, \qquad S' = (S \setminus \{2\}) \cup \{h + 1\}.$$

Both truth-one weights are $1/2$. Atom 1 belongs to each truth-one set and to neither truth-zero set, giving the gap $1/h$. The readout $\{1, 2\}$ contains two atoms of S and one of S'. Finally a semantic gauge can take S only to itself or its complement; S' is neither, since it omits 2 but contains 1. □

Thus additional unlabeled output coverage cannot resolve this family: the obstruction survives observation of the complete joint law. For each fixed h, it also occurs at positive weights and unit singular values. Information that distinguishes the scientific grouping must enter through a stronger semantic predicate.

4.2. **The full diagonal fiber.** Fix a connected design, $Y_g = J = \{1, \dots, r\}$ at every group, and positive p_j with $\sum_j p_j = 1$. On each edge let

$$P_e^{\mathrm{diag}}(x) = \begin{cases} p_j, & x_g = j \text{ for every } g \in e, \\ 0, & \text{otherwise.} \end{cases} \tag{4.1}$$

The next result starts with all real competitors in Ξ_r, rather than a predetermined atomic submodel.

Theorem 4.3 (Exact diagonal observational fiber)**.** *The fiber $\mathcal{F}(P^{\mathrm{diag}}, \mathrm{true})$ consists exactly of parameters indexed by bijections $b : Z \to J$ with*

$$w(z) = p_{b(z)}, \qquad L_g(y, z) = \mathbf{1}\{y = b(z)\} \quad (g \in G). \tag{4.2}$$

In particular, deterministic kernels and full column rank are consequences of the observed law.

Proof. Take an arbitrary parameter in the fiber. Fix an edge and a latent state z. Let the three nonempty column supports be A_1, A_2, A_3. At each off-diagonal observation cell the sum of nonnegative mixture terms is zero. Strict positivity of the z weight implies that its product term vanishes there. Therefore

$$A_1 \times A_2 \times A_3 \subseteq \{(j, j, j) : j \in J\}.$$

Choose one element from each support. They have to equal some j. A second element in any support, paired with the other two chosen elements, would produce a supported off-diagonal cell. Hence each support is the same singleton, and normalization makes its column deterministic.

Write $b_e(z)$ for the resulting atom on edge e. Shared group kernels imply $b_e(z) = b_f(z)$ on adjacent edges; connectedness gives one map $b : Z \to J$. At a diagonal cell,

$$p_j = \sum_{z:b(z)=j} w(z).$$

Since every $p_j > 0$, b is surjective. Both its domain and codomain have r elements, so b is bijective, and the last display gives $w(z) = p_{b(z)}$. Conversely, every parameter in (4.2) has the required observations. □

The argument displays where fixed state count matters. With more than r positive latent terms, an observed atom could be split among several terms; that is a different comparison model. The uniqueness conclusion also follows from the classical tensor theorem and synchronization. The elementary support proof makes the exact domain reduction explicit for the semantic calculation.

4.3. **Attained sharp semantic images.** Let $k = m_1 \in \{1, \dots, r-1\}$. For a k-element subset $S \subseteq J$, put

$$u(S) = \sum_{j \in S} p_j, \tag{4.3}$$

$$a_B(S) = \frac{\sum_{j \in S \cap B} p_j}{u(S)} - \frac{\sum_{j \in (J \setminus S) \cap B} p_j}{1 - u(S)}, \tag{4.4}$$

$$f_R(S) = \frac{\sum_{j \in S \cap R} p_j}{u(S)}, \qquad s_R(S) = \frac{\sum_{j \in (J \setminus S) \cap (J \setminus R)} p_j}{1 - u(S)}. \tag{4.5}$$

All denominators lie strictly between zero and one. For a fixed finite anchor family $\mathcal{A}$, let

$$\mathcal{S}_{\mathcal{A}} = \{S \subseteq J : |S| = k,\ s\, a_B(S) > 0 \text{ for every } (g, B, s) \in \mathcal{A}\}.$$

Theorem 4.4 (Sharp diagonal performance images)**.** *For the full real fiber in Theorem 4.3, the sensitivity, specificity, and truth-prevalence images are, respectively,*

$$\{f_R(S) : S \in \mathcal{S}_{\mathcal{A}}\}, \qquad \{s_R(S) : S \in \mathcal{S}_{\mathcal{A}}\}, \qquad \{u(S) : S \in \mathcal{S}_{\mathcal{A}}\}. \tag{4.6}$$

The anchored fiber is empty exactly when $\mathcal{S}_{\mathcal{A}}$ is empty. Otherwise each displayed image has an attained minimum and maximum.

Proof. For a bijection b in (4.2), the observable atoms assigned to truth one are $S = b(\tau^{-1}(1))$, with $|S| = k$. Substitution into (2.5)–(2.7) gives (4.3)–(4.5), and the anchors are precisely the defining inequalities of $\mathcal{S}_{\mathcal{A}}$. Conversely, every k-element subset is obtained by choosing bijections from the two truth blocks to S and its complement. This proves both inclusions in each image equality and the emptiness statement. Finiteness gives attained extrema and the same bijections provide attainment witnesses. □

The image formulas are finite mathematical descriptions for arbitrary real p_j. An executable exact enumeration additionally requires a representation that decides the displayed equalities and strict inequalities, for example rational or suitably encoded real-algebraic masses. Finiteness and attainment alone do not provide those decisions for unspecified real inputs.

Example 4.5 (A four-state hidden-factor model)**.** Take one edge $\{a, b, c\}$, $Z = \{0,1\} \times \{0,1\}$, $\tau(t,h) = t$, and $p_j = 1/4$ for $j = 1,2,3,4$. Two admissible atom assignments are

(t,h)	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$b(t,h)$	3	4	1	2
$b'(t,h)$	2	4	1	3

At each group use the deterministic kernel associated with the displayed assignment. Both worlds have the diagonal observation $P^*(j,j,j) = 1/4$, the fixed anchor $(a, \{1\}, +1)$ has gap $1/2$, and the readout is $R = \{1,2\}$. The possible anchored truth-one sets are exactly

$$\{1,2\}, \quad \{1,3\}, \quad \{1,4\}.$$

Consequently the full sensitivity image, and also the full specificity image, is $\{1/2, 1\}$; prevalence is $1/2$. Their sensitivity and specificity interval hulls have width $1/2$. Theorem 4.3 proves that every full-model competitor is one of the equal-weight deterministic bijection models. Thus the unrestricted fiber, its rank-four restriction, and its atomic restriction coincide at P^*, including after the anchor is imposed.

Example 4.6 (One additional state changes the semantic image)**.** Keep the four observed atoms and equal masses of Example 4.5, but compare five positive latent states with truth-block sizes $(m_0, m_1) = (3,2)$. For $0 < \alpha < 1/4$, assign atom 1 mass $1/4$ in truth one; split atom 2 into mass α in truth one and mass $1/4 - \alpha$ in truth zero; assign atoms 3 and 4 mass $1/4$ each in truth zero. Every group kernel is deterministic at its assigned atom. The full observed law is unchanged, while

$$D_{a,\{1\},+1} = \frac{1/4}{1/4 + \alpha} > 0, \qquad s_{g,\{1,2\}} = \frac{1/2}{3/4 - \alpha}.$$

The latter ranges over $(2/3, 1)$. Thus a continuous family of semantic values is possible after a single atom is split. This changes both the comparator state count and its truth-block profile; it is outside the fixed-r conclusion of Theorem 4.4.

5. Explicit conditioning and inverse bounds

5.1. **Coordinates, norms, and the conditioned class.** Embed the parameter arrays in Euclidean space, counting each component weight and each group kernel entry exactly once. Its dimension is

$$p = |\mathcal{C}|r + r \sum_{g \in G} n_g.$$

Let A_0 be the affine space defined by all normalization equalities in (2.1), and V its parallel linear space. Thus tangent weights and each tangent kernel column sum to zero. The polynomial formula (2.2) extends to all of A_0. Observation norms are the Euclidean norm of all stacked edge cells, with

$$m = \sum_{e \in \mathcal{E}} d_e.$$

The permutation action is an isometry. For any $\Gamma \leq W$, put

$$d_\Gamma(\theta, \eta) = \min_{q \in \Gamma} \|\theta - S_q \eta\|_2 .$$

This is a pseudometric on arrays and a metric on orbits: the minimum is attained, zero means orbit equality, and the triangle inequality follows by composing minimizing isometries.

Assume $n_g \geq r$ and supply margins

$$0 < w_0 \leq 1/r, \qquad 0 < \sigma_0 \leq 1. \tag{5.1}$$

Define

$$\mathcal{K} = \{\theta \in \Xi_r : w_c(z) \geq w_0 \text{ for every } c, z, \quad L_g^{\mathsf{T}} L_g - \sigma_0^2 I_r \succeq 0 \text{ for every } g\}. \tag{5.2}$$

The matrices have smallest singular value at least σ_0. Zero kernel entries remain allowed.

Lemma 5.1. *The set $\mathcal{K}$ is nonempty, compact, and W-invariant. For $\theta, \eta \in \mathcal{K}$, observation equality is equivalent to $d_W(\theta, \eta) = 0$.*

Proof. Use equal weights $1/r$ and embed the r coordinate vectors as the first r outcome columns at each group. The Gram matrices are I_r, so this parameter belongs to $\mathcal{K}$. The normalization, nonnegativity, weight-margin, and Gram inequalities are closed conditions; $w_0 > 0$ ensures they remain inside Ξ_r. All entries lie in $[0, 1]$, giving compactness. A column permutation conjugates the Gram matrix by an orthogonal permutation matrix, proving invariance. Full column rank implies (3.5), and Theorem 3.4 proves the final assertion. Appendix A gives a direct full-rank proof. □

Define the following constants solely from the design and supplied margins:

$$a_g = \frac{1 + \sqrt{n_g}}{w_0 \sigma_0^2}, \qquad \beta_e = \sigma_0^{-3} + \sigma_0^{-1} \sum_{g \in e} a_g, \tag{5.3}$$

$$B = \max_{e \in \mathcal{E}} \left(\beta_e^2 + \sum_{g \in e} a_g^2 \right)^{1/2}, \qquad \kappa = B^{-1}, \tag{5.4}$$

$$M = 12\sqrt{m}, \qquad \rho = \min\{1, \kappa/M\}, \tag{5.5}$$

$$D = \sqrt{2(|\mathcal{C}| + r|G|)}. \tag{5.6}$$

All are finite and positive.

Theorem 5.2 (Uniform derivative and diameter bounds)**.** *For every $\theta \in \mathcal{K}$ and $v \in V$,*

$$\|D\mathcal{O}_\theta(v)\|_2 \geq \kappa \|v\|_2 . \tag{5.7}$$

For every $x \in A_0$ with $\|x - \theta\|_2 \leq 1$ and every $u, v \in V$,

$$\left\|D^2 \mathcal{O}_x[u, v]\right\|_2 \leq M \|u\|_2 \|v\|_2 . \tag{5.8}$$

Moreover, $d_W(\theta, \eta) \leq D$ for all $\theta, \eta \in \mathcal{K}$.

Proof. For each group choose a Euclidean left inverse Λ_g satisfying

$$\Lambda_g L_g = I_r, \qquad \|\Lambda_g\|_{\mathrm{op}} \leq \sigma_0^{-1}.$$

It is obtained by inverting L_g on its column space and setting it to zero on the orthogonal complement. Mode contraction of a tensor by Λ_g multiplies its Frobenius norm by at most σ_0^{-1}.

Fix an edge $e = \{a, b, c\}$ and write $J_e = D\mathcal{O}_{\theta,e}(v)$. Let $\dot{w}$ and $\dot{L}_g$ denote its tangent arrays. Contract J_e in the b, c modes by Λ_b, Λ_c and let t_z be the remaining vector at latent slice (z, z). Differentiation gives

$$t_z = w_z \dot{l}_{a,z} + \alpha_z l_{a,z}, \qquad \alpha_z = \dot{w}_z + w_z(\Lambda_b \dot{L}_b)_{zz} + w_z(\Lambda_c \dot{L}_c)_{zz}.$$

Normalization tangency and $\mathbf{1}^\mathsf{T} l_{a,z} = 1$ imply $\alpha_z = \mathbf{1}^\mathsf{T} t_z$. Therefore

$$\dot{l}_{a,z} = \frac{t_z - l_{a,z}(\mathbf{1}^\mathsf{T} t_z)}{w_z}, \qquad \left\|\dot{l}_{a,z}\right\|_2 \le \frac{1+\sqrt{n_a}}{w_0} \|t_z\|_2 .$$

The selected diagonal slices satisfy $\sum_z \|t_z\|_2^2 \le \sigma_0^{-4} \|J_e\|_2^2$. Hence

$$\left\|\dot{L}_g\right\|_F \le a_g \|J_e\|_2 \qquad (g \in e). \tag{5.9}$$

Contracting all three modes and retaining the fully diagonal vector d gives

$$d_z = \dot{w}_z + w_z \sum_{g \in e} (\Lambda_g \dot{L}_g)_{zz}.$$

Diagonal selection is norm decreasing and $w_z \le 1$, so

$$\|\dot{w}\|_2 \le \sigma_0^{-3} \|J_e\|_2 + \sigma_0^{-1} \sum_{g \in e} \left\|\dot{L}_g\right\|_F \le \beta_e \|J_e\|_2 . \tag{5.10}$$

Squaring the local bounds and summing over edges yields

$$\|v\|_2^2 \le \sum_{e \in \mathcal{E}} \left(\left\|\dot{w}_{[e]}\right\|_2^2 + \sum_{g \in e} \left\|\dot{L}_g\right\|_F^2 \right) \le B^2 \sum_{e \in \mathcal{E}} \|J_e\|_2^2 .$$

The first inequality uses coverage: every global parameter block appears at least once, while the global norm counts it once. This proves (5.7) without inserting edge multiplicities into the norm.

For the Hessian bound, every coordinate of x has absolute value at most two, since θ has coordinates in $[0,1]$ and $\|x - \theta\|_2 \le 1$. Each scalar observation polynomial is a sum over z of products of four distinct coordinates, one weight and three kernel entries. Its ambient Hessian is block diagonal in z. Within each four-coordinate block, a row has three off-diagonal entries, each a product of two remaining coordinates and therefore of absolute value at most four. Its absolute row and column sums are at most twelve. Thus its bilinear operator norm is at most twelve. Squaring and summing over the m observation cells gives (5.8). This argument applies on the entire affine ball, including its nonstochastic points.

Finally two probability vectors have squared Euclidean distance at most two. A parameter contains $|\mathcal{C}|$ weight vectors and $r|G|$ stochastic kernel columns. Their sum gives the direct distance bound D, and the quotient distance is no larger. □

5.2. **Local and global inversion.**

Proposition 5.3 (Local inverse)**.** *For $\theta, \eta \in \mathcal{K}$ with $d_W(\theta, \eta) \le \rho$,*

$$d_W(\theta, \eta) \le \frac{2}{\kappa} \|\mathcal{O}(\theta) - \mathcal{O}(\eta)\|_2 . \tag{5.11}$$

Proof. Choose q attaining the orbit distance and set $v = S_q \eta - \theta \in V$. By invariance, $\mathcal{O}(S_q \eta) = \mathcal{O}(\eta)$. Taylor's integral formula and Theorem 5.2 imply

$$\|\mathcal{O}(\eta) - \mathcal{O}(\theta)\|_2 \ge \kappa \|v\|_2 - \frac{M}{2} \|v\|_2^2 \ge \frac{\kappa}{2} \|v\|_2 ,$$

using $\rho \le 1$ and $M\rho \le \kappa$. The line segment is in A_0 and within the unit ball where the Hessian estimate holds. □

Define the far-pair set

$$Q_\rho = \{(\theta,\eta) \in \mathcal{K}^2 : d_W(\theta,\eta) \geq \rho\}. \tag{5.12}$$

It is compact. If it is nonempty, the continuous function $\|\mathcal{O}(\theta) - \mathcal{O}(\eta)\|_2^2$ attains a strictly positive minimum there: a zero would contradict Lemma 5.1 and $\rho > 0$. If it is empty, any positive lower bound is valid for its defining implication. The next section gives a concrete terminating program that computes a valid positive bound without calculating this minimum.

Corollary 5.4 (Global inverse from a separation certificate)**.** *Suppose $\Delta > 0$ satisfies*

$$d_W(\theta,\eta) \geq \rho \quad \Longrightarrow \quad \|\mathcal{O}(\theta) - \mathcal{O}(\eta)\|_2 \geq \Delta \qquad (\theta,\eta \in \mathcal{K}). \tag{5.13}$$

Then, for every $\theta, \eta \in \mathcal{K}$,

$$d_W(\theta,\eta) \leq C_{\mathrm{inv}} \|\mathcal{O}(\theta) - \mathcal{O}(\eta)\|_2, \qquad C_{\mathrm{inv}} = \max\{2/\kappa, D/\Delta\}. \tag{5.14}$$

If the observation distance is strictly less than Δ, the coefficient $2/\kappa$ applies.

Proof. For $d_W \leq \rho$ use Proposition 5.3. For $d_W \geq \rho$, combine (5.13) with $d_W \leq D$. The strict observation threshold excludes the far-pair case and gives localization. □

6. A certified positive separation algorithm

The compactness argument preceding Corollary 5.4 establishes existence of a separation bound. This section makes a positive bound computable and its validity independently checkable. The output is a positive reciprocal power of two and a finite cover certificate. Both the computation and its proof concern the full real set $\mathcal{K}$. The result is computability in principle: the certificate-size bound below precludes practical explicit trees in some elementary parameter regimes.

6.1. Exact inputs and algebraic margins.

The combinatorial input is a finite ordered group list, a nonempty list of distinct triple edges, alphabet sizes $n_g \geq r$, and $r \geq 2$. A validator checks coverage and these constraints and computes the edge components. The margins in (5.1) are supplied by finite real-algebraic codes of either form:

(1) a rational number a;
(2) an integer polynomial p of positive degree and a rational bracket $[l,u]$ with $l < u$, $p(l)p(u) < 0$, and a rational Horner interval for p' on $[l,u]$ excluding zero.

The second code denotes the unique root in (l,u). Indeed, the intermediate-value theorem gives existence and the derivative sign gives strict monotonicity. Appendix B proves adequacy for every real algebraic number and the required exact refinement and rational comparison procedures. Input validity includes the comparisons $0 < w_0 \leq 1/r$ and $0 < \sigma_0 \leq 1$. No model parameter is required to be algebraic; finite coding is imposed only on the supplied margins.

6.2. An exact full-real polynomial encoding.

Let

$$d = |\mathcal{C}|(r-1) + r\sum_{g\in G}(n_g - 1), \qquad \nu = 2d. \tag{6.1}$$

Store all but the last entry of each probability vector. Reconstruct its last entry as one minus the sum of the stored entries. This gives an affine bijection $A : \mathbb{R}^d \to A_0$. For a pair use $x = (u,v) \in X = [0,1]^\nu$, with $\theta = A(u)$ and $\eta = A(v)$.

Generate a fixed finite list of polynomial constraints $g_j(x) \geq 0$ consisting of:

(1) every weight entry, stored or reconstructed, minus w_0, for both pair members;
(2) every kernel entry, stored or reconstructed, for both pair members;

(3) every nonempty principal minor of $L_g^{\mathsf{T}} L_g - \sigma_0^2 I_r$, for both pair members and every group;
(4) the polynomials

$$F_q(x) = \|A(u) - S_q A(v)\|_2^2 - \rho^2 \qquad (q \in W). \tag{6.2}$$

The entire permutation set W is generated from the design. The full norm in (6.2) is evaluated after reconstruction. Define

$$Q = \{x \in X : g_j(x) \geq 0 \text{ for every } j\}, \qquad H(x) = \|\mathcal{O}(A(u)) - \mathcal{O}(A(v))\|_2^2 . \tag{6.3}$$

The polynomial H has degree at most eight. All expressions use rational constants, variables, addition, negation, multiplication, and the three exact scalar leaves w_0, σ_0^2, ρ^2.

Lemma 6.1 (Exactness of the encoding)**.** *The map $(u, v) \mapsto (A(u), A(v))$ is a bijection from Q onto Q_ρ, and H is exactly the squared observation distance on this set.*

Proof. Normalization elimination is an affine bijection, and every stochastic vector has its stored entries in $[0, 1]$. All entries, including the reconstructed ones, are tested in the constraints. For a real symmetric matrix, positive semidefiniteness is equivalent to nonnegativity of all principal minors; this includes singular boundaries. The finite conjunction over all $q \in W$ is equivalent to the minimum orbit distance being at least ρ. These observations give both directions and uniqueness of the encoding. The objective equality follows from its definition. □

6.3. **Interval evaluation and certificates.** Use closed intervals with exact rational endpoints. Addition and negation have their usual endpoint rules; multiplication takes the minimum and maximum of the four endpoint products. Horner or syntax-tree evaluation then encloses the polynomial on a rational box. Repeated scalar occurrences denote one exact scalar, even if interval evaluation encloses those occurrences separately.

At precision $s \geq 0$, refine every margin currently represented by a Root code to width at most 2^{-s-2}, unless an exact midpoint root changes it to a Rat code. Rat codes use singleton enclosures. Evaluate (5.3)–(5.5) with outward rational interval arithmetic. Reciprocals are taken only on positive intervals. For square roots use outward dyadic endpoints on mesh 2^{-s-2}, computed by integer square-root comparisons; finite minima and maxima use endpoint minima and maxima. Positive preparation and the explicit coarse-stage bounds in Appendix B give valid denominators at every stage. The resulting scalar enclosures contain the exact coefficients and their widths tend to zero. No rounded value replaces the radius ρ in the mathematical constraints.

A certificate contains the exact input, a format identifier, an integer $k \geq 0$, checked scalar refinements and precision, and a finite binary tree. The root box is X. An internal node specifies a coordinate and includes both closed midpoint halves of its parent. The checker reconstructs these boxes; they are not trusted input. A leaf has one of two reasons:

Exclusion: A constraint index j whose interval upper endpoint is strictly negative.
Gap: The interval lower endpoint of H is at least 2^{-2k}.

The checker regenerates the affine map, full permutation family, constraints and objective. It checks scalar refinements against the original codes and re-evaluates every leaf. Missing children, malformed indices, invalid scalar data, or unjustified leaf reasons reject. Accepted certificates return $\Delta = 2^{-k}$.

Proposition 6.2 (Certificate soundness)**.** *For every admitted input and every finite certificate accepted by this checker, $\Delta > 0$ and (5.13) holds. If all leaves are exclusions, Q_ρ is empty.*

Proof. The two closed midpoint halves cover their parent. Induction on the finite tree shows that the leaf boxes cover X, including shared boundaries. Interval inclusion shows that an exclusion box contains no point of Q. On a gap box, $H \geq 2^{-2k}$. Every point of Q belongs to a leaf and cannot belong to an exclusion leaf; therefore it satisfies the gap bound. Apply Lemma 6.1 and take nonnegative square roots. If every leaf is an exclusion, no feasible point exists. □

Soundness is independent of the certificate generator and does not assume the existence of a positive gap. A tree containing gap leaves alone gives no conclusion about whether the far-pair set is empty.

6.4. **The terminating search.** At stage $s = 0, 1, 2, \ldots$, set $k = s$ and use the stage-s coefficient enclosures. Traverse boxes depth first in a fixed order. At a box, test exclusions first and then the gap condition. If neither succeeds and the depth is νs, declare the stage inconclusive. Otherwise bisect coordinate depth mod ν, visit both children, and retain a complete node only if both succeed. A failing branch may abort the stage. Replay a successful tree with the checker and return only after acceptance.

Each stage has at most $2^{\nu s+1} - 1$ generator visits. On a full-depth unpruned path every coordinate has been split exactly s times, giving side length 2^{-s}. Early accepted leaves may be larger; their checked inequalities justify them directly.

Lemma 6.3 (Uniform interval convergence)**.** *For the finite family $g_1, \ldots, g_t, H$, the interval endpoint errors tend uniformly to zero as coefficient precision tends to infinity and the maximum box side length tends to zero. In particular this holds along the stage schedule just described.*

Proof. The scalar refinements enclose fixed exact values and their widths tend to zero. All scalar intervals are eventually in one bounded neighborhood of those values. For a variable, endpoint error on a box is at most its side length. Constants and scalar leaves satisfy the corresponding bounds. Addition and negation preserve convergence. For multiplication, boundedness gives

$$|ab - a'b'| \leq |a|\,|b - b'| + |b'|\,|a - a'|.$$

Structural induction over each finite expression gives a uniform modulus on the compact coordinate box and bounded coefficient neighborhood. Take the maximum over the finitely many expressions. Details for the square-root scalar construction appear in Appendix B. □

Theorem 6.4 (Computed positive separation)**.** *For every covered finite triple design and every admitted pair of algebraic margins satisfying* (5.1)*, the specified rational search terminates with a finite accepted certificate and a positive $\Delta = 2^{-k}$ satisfying* (5.13)*. In fact, all sufficiently late stages succeed.*

Proof. Each fixed stage terminates: it has a finite depth bound, finite expression lists, finite rational arithmetic, and terminating scalar refinement. By Lemma 5.1 and Lemma 6.1, $H > 0$ on Q, and Q is compact. If Q is nonempty choose a reciprocal dyadic $\delta_0 > 0$ with $\delta_0^2 < \min_Q H$. If Q is empty choose $\delta_0 = 1$. These are proof witnesses; the program is not given them.

On all of X, the continuous function

$$\phi(x) = \max\{H(x) - \delta_0^2, -g_1(x), \ldots, -g_t(x)\}$$

is strictly positive. At a feasible point its first term is positive; at an infeasible point some constraint is negative. Compactness gives a positive minimum $a = \min_X \phi$.

By Lemma 6.3, for all sufficiently large s, all endpoint errors on boxes of side length at most 2^{-s} are less than $a/2$. Increase s further so that $2^{-s} \leq \delta_0$. At any such box choose a point x. Either a constraint has $-g_j(x) \geq a$, in which case its interval upper endpoint is negative,

or $H(x) - \delta_0^2 \geq a$, in which case its interval lower endpoint exceeds $\delta_0^2 \geq 2^{-2s}$. Thus every full-depth box admits a passing label. Earlier nodes either pass directly or have successful children. Induction upward gives success of the actual stage and agreement with checker replay. Proposition 6.2 gives the returned separation bound. □

The proof uses a fixed positive slack before comparing the stage threshold with it; no rate such as interval error $o(4^{-s})$ is required. The returned value need not be the best valid dyadic and need not equal the square root of the far-pair minimum.

The implementation has both an unbounded successful-value program and a finite-fuel interface. Insufficient fuel returns an inconclusive status, and a positive result is published only after complete serialized-certificate replay. Finite-fuel termination and prefix stability are formal properties of the program. Physical CPU, memory, wall-clock, and allocation bounds belong to an execution supervisor and are distinct from this mathematical termination theorem.

6.5. **An output-size obstruction for the box certificate.** Let $\mathcal{K}_u = A^{-1}(\mathcal{K}) \subseteq [0,1]^d$, with the Euclidean norm on $\mathbb{R}^d$, and let vol_d denote Lebesgue volume in the stored coordinates. The following lower bound concerns the explicit tree and the two leaf rules specified above. It does not assume any particular search order.

Proposition 6.5 (Leaf-count lower bound)**.** *Every accepted certificate with L leaves satisfies*

$$L > \mathrm{vol}_d(\mathcal{K}_u) \left(\frac{\sqrt{d}}{2\rho} \right)^d . \tag{6.4}$$

This statement uses axis-aligned product boxes $U \times V$ in the original pair coordinates, exclusion by a single compiled constraint with a strictly negative interval upper endpoint, and a strictly positive gap threshold for H.

Proof. The admitted inputs give $d \geq 1$ and $\rho > 0$, and Lemma 5.1 makes $\mathcal{K}_u$ nonempty. For every $u \in \mathcal{K}_u$, the diagonal point (u, u) belongs to the root box and hence to some accepted leaf. At that point all model constraints are nonnegative and $H(u,u) = 0$. Thus no model constraint can exclude this leaf, and it cannot be a gap leaf. Its accepting reason must be a distance constraint F_q.

Consider any such leaf $U \times V$ and fix $v_0 \in V$. Enclosure and the strict exclusion test give

$$\|A(x) - S_q A(v_0)\|_2 < \rho \qquad (x \in U).$$

Consequently, for $x, y \in U$,

$$\|x - y\|_2 \leq \|A(x) - A(y)\|_2 < 2\rho.$$

The first inequality holds because the stored coordinates occur unchanged among the full reconstructed entries. If $\ell_1, \ldots, \ell_d$ are the side lengths of U, apply this inequality to opposite corners and then the arithmetic–geometric mean inequality to obtain

$$\sum_{i=1}^{d} \ell_i^2 < 4\rho^2, \qquad \mathrm{vol}_d(U) = \prod_{i=1}^{d} \ell_i < \left(\frac{2\rho}{\sqrt{d}} \right)^d .$$

This remains true if a side length is zero. The projections of the finitely many leaves meeting the admissible diagonal cover $\mathcal{K}_u$. Its compactness ensures measurability, so finite subadditivity gives

$$\mathrm{vol}_d(\mathcal{K}_u) \leq \sum_{\text{diagonal leaves}} \mathrm{vol}_d(U) < L \left(\frac{2\rho}{\sqrt{d}} \right)^d .$$

Shared boundaries and overlaps of the projected boxes do not affect this argument. Rearranging proves the claim. □

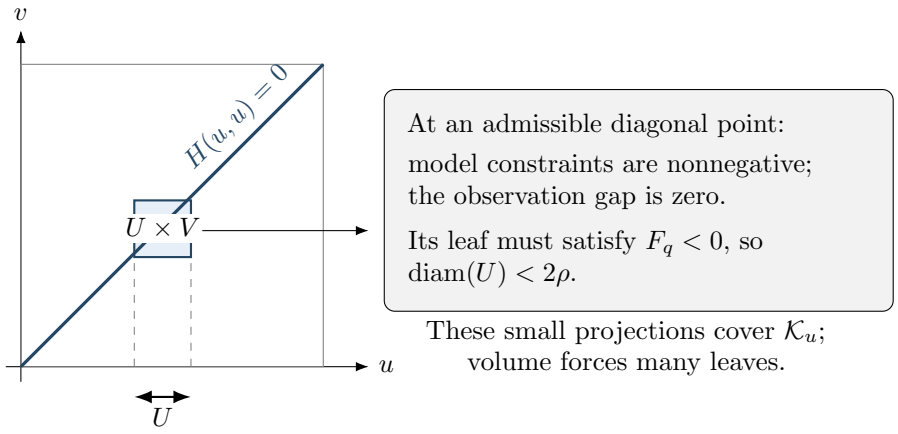


FIGURE 2. Why an infeasible diagonal controls certificate size. The picture is a one-coordinate schematic of a pair box; the proof uses d-dimensional projections. The root contains every (u, u) with $u \in \mathcal{K}_u$, even though none is a far pair.

Example 6.6 (The binary one-edge scale)**.** For $r = n_g = 2$ on one triple, $d = 7$ and $m = 8$. Suppose $w_0 < 1/2$ and $\sigma_0 < 1$. A kernel near the identity can be written

$$L = \begin{pmatrix} 1-a & b \\ a & 1-b \end{pmatrix}, \qquad a, b \geq 0.$$

Its perturbation from I_2 has operator norm $\sqrt{2(a^2+b^2)}$. Therefore the quarter disk $a^2 + b^2 \leq (1-\sigma_0)^2/2$ is admissible. A second quarter disk lies at the swap matrix; their interiors are disjoint. Their combined area in the two stored kernel coordinates is $\pi(1-\sigma_0)^2/4$. The weight interval and three independent kernels give

$$\operatorname{vol}_7(\mathcal{K}_u) \geq (1-2w_0)\left(\frac{\pi(1-\sigma_0)^2}{4}\right)^3. \tag{6.5}$$

Here $a_g = (1+\sqrt{2})/(w_0\sigma_0^2)$, $\beta_e = \sigma_0^{-3} + 3a_g/\sigma_0$, and $\rho = 1/(12\sqrt{8}\sqrt{\beta_e^2 + 3a_g^2})$. Substitution into (6.4) gives the following scales; the displayed decimals approximate the exact analytic lower expressions.

w_0	σ_0	Lower expression for L (approximately)
$1/5$	$1/5$	1.3804×10^{36}
$1/4$	$1/\sqrt{2}$	2.9369×10^{21}
$49/100$	$9/10$	1.6710×10^{13}

For example, rational enclosures of the radicals and π give the strict bound $L > 2.9369 \times 10^{21}$ in the middle row. The exact expressions (6.4)–(6.5), rather than rounded table entries, define the bounds.

The obstruction is specific to this certificate language. Pruning, branch ordering, or search reuse cannot make its final explicit tree smaller than (6.4); a quotient-domain checker, a compressed symmetry certificate, different coordinates, or stronger relational leaf rules would

require a different analysis. The bound is vacuous when $\mathrm{vol}_d(\mathcal{K}_u) = 0$, and interior margins alone do not ensure that it is numerically large. In particular, it does not explain the difficulty of a finite boundary model such as Example 7.5. It is compatible with the termination theorem.

There is also a separate representation cost. In the accompanying executable serialization, a natural token n is encoded by n unit bytes followed by a terminator. A denominator 2^t therefore requires $2^t + 1$ bytes for that token alone, rather than a number of bits proportional to t. This unary encoding is total and admits exact replay, but is not a compact binary encoding. Replacing it would address representation size without removing the geometric lower bound above.

7. Statistical consequences and semantic information

7.1. **Finite-block sampling and subclass error.** For each edge, fix an integer $N_e \geq 1$ and a probability law P_e on X_e. The sample space is

$$\Omega_O = \prod_{e \in \mathcal{E}} X_e^{N_e},$$

with its full power-set sigma algebra. Let μ be any probability law whose entire e-block marginal equals $P_e^{\otimes N_e}$. Distinct blocks may have arbitrary dependence. Let $\widehat{P}_e$ be the empirical frequency vector, including all cells, and stack these vectors into $\widehat{P}$.

For $0 < \delta < 1$, set

$$t_e(\delta) = \sqrt{\frac{\log(2m/\delta)}{2N_e}}, \qquad s_\delta = \left(\sum_{e \in \mathcal{E}} \frac{d_e \log(2m/\delta)}{2N_e} \right)^{1/2}. \tag{7.1}$$

The Bernoulli Hoeffding bound and a union bound over all m cells give an event of probability at least $1 - \delta$ on which every cell error is at most $t_e(\delta)$ and hence

$$\left\| \widehat{P} - P \right\|_2 \leq s_\delta. \tag{7.2}$$

Only within-block product laws enter the cellwise tail argument. No independence between edges is used.

Fix $\theta_* \in \mathcal{K}$ and deterministic $\xi, \epsilon \geq 0$ with

$$\|P - \mathcal{O}(\theta_*)\|_2 \leq \xi.$$

Let $\widehat{\theta} : \Omega_O \to \mathcal{K}$ be a deterministic estimator satisfying the following all-outcome, all-competitor condition:

$$\left\| \mathcal{O}(\widehat{\theta}(\omega)) - \widehat{P}(\omega) \right\|_2 \leq \left\| \mathcal{O}(\zeta) - \widehat{P}(\omega) \right\|_2 + \epsilon \quad \text{for every } \omega \in \Omega_O,\ \zeta \in \mathcal{K}. \tag{7.3}$$

This is an additive near-global fit guarantee. Membership in $\mathcal{K}$, stationarity, or a small gradient alone does not establish (7.3).

Theorem 7.1 (Finite-sample subclass rate)**.** *Let $\Delta > 0$ satisfy* (5.13)*, and define*

$$E_\delta = 2(\xi + s_\delta) + \epsilon.$$

With probability at least $1 - \delta$, simultaneously,

$$\left\| \mathcal{O}(\widehat{\theta}) - \mathcal{O}(\theta_*) \right\|_2 \leq E_\delta, \tag{7.4}$$

$$d_W(\widehat{\theta}, \theta_*) \leq \min\{D, C_{\mathrm{inv}} E_\delta\}. \tag{7.5}$$

If $E_\delta < \Delta$, the same event also satisfies

$$d_W(\widehat{\theta}, \theta_*) \le \min\left\{D, \frac{2E_\delta}{\kappa}\right\} = \min\left\{D, \frac{4}{\kappa}(\xi + s_\delta) + \frac{2\epsilon}{\kappa}\right\}. \tag{7.6}$$

Proof. On (7.2), $\left\|\widehat{P} - \mathcal{O}(\theta_*)\right\|_2 \le \xi + s_\delta$. Compare (7.3) to $\zeta = \theta_*$ and use the triangle inequality once more to obtain (7.4). Corollary 5.4 and the diameter bound give the remaining statements on that same event. □

Remark 7.2 (A blockwise concentration refinement)*.* The inversion proof uses only a valid bound on $\left\|\widehat{P} - P\right\|_2$. A less dimension-dependent choice is

$$s_\delta^{\mathrm{vec}} = \left[\sum_{e\in\mathcal{E}} \frac{\left(1 + \sqrt{\log(|\mathcal{E}|/\delta)}\right)^2}{N_e}\right]^{1/2}.$$

Indeed, for one empirical categorical law, $\mathbb{E}\left\|\widehat{P}_e - P_e\right\|_2^2 = (1 - \|P_e\|_2^2)/N_e \le 1/N_e$, so Jensen's inequality bounds the mean norm by $N_e^{-1/2}$. Replacing one observation changes this norm by at most $\sqrt{2}/N_e$. The bounded-differences inequality [15] gives an upper deviation $\sqrt{\log(|\mathcal{E}|/\delta)/N_e}$ with failure probability at most $\delta/|\mathcal{E}|$. A union bound over edges proves the displayed stacked radius without cross-edge independence. Replacing s_δ by s_δ^{vec} in Theorem 7.1 is therefore valid. This refinement leaves the potentially conservative inverse constants unchanged; the particular sample prescription below uses (7.1).

Theorem 6.4 provides one way to obtain such a Δ from algebraic margins; an analytic separation proof is equally sufficient. For fixed design and margins, the sampling term has order $N_{\min}^{-1/2}$. Persistent model discrepancy or optimization error gives the displayed nonvanishing floor. For normalized laws p, q on the same finite alphabet, define $\mathrm{TV}(p,q) = \frac{1}{2}\|p - q\|_1$, so $0 \le \mathrm{TV}(p,q) \le 1$. If deterministic edgewise bounds $\mathrm{TV}(P_e, \mathcal{O}(\theta_*)_e) \le b_e$ are independently justified, then

$$\xi = 2\left(\sum_e b_e^2\right)^{1/2} \tag{7.7}$$

is valid because $\|v\|_2 \le \|v\|_1$.

A deterministic exact baseline satisfying (7.3) with $\epsilon = 0$ exists. Minimize the continuous squared residual over nonempty compact $\mathcal{K}$ and select the lexicographically least minimizer in a fixed coordinate order. Successive coordinate minimization over compact minimizer sets gives a unique selected vector for every sample. Since the sample space is finite, this defines a measurable estimator. This existence result is sufficient for the next mathematical corollary; an executable globally certified fitting implementation is a separate task.

Corollary 7.3 (A finite informative regime)**.** *For any conditioned design and any $\Delta > 0$ satisfying* (5.13)*, put*

$$a = \min\{\kappa D/8, \Delta/4\}, \qquad N = \left\lceil 20m^2/a^2 \right\rceil.$$

Use the identity-column parameter from Lemma 5.1 as θ_*, let $P = \mathcal{O}(\theta_*)$, take $N_e = N$ for all edges, and use the exact baseline estimator. With probability at least $19/20$, its error is at most $D/2$, and the strict small-error condition in Theorem 7.1 holds.

Proof. For $\delta = 1/20$,

$$s_\delta^2 = \frac{m \log(40m)}{2N} \le a^2 \frac{\log(40m)}{40m} < a^2.$$

Thus $E_\delta = 2s_\delta < 2a \le \Delta/2 < \Delta$, and $2E_\delta/\kappa < 4a/\kappa \le D/2$. Apply Theorem 7.1. □

This corollary establishes a finite regime below the universal diameter using explicitly specified constants. It does not select an application-specific sample budget.

Example 7.4 (Conservatism of the prescribed sample budget)**.** Consider one triple with $r = n_g = 2$ and margins $w_0 = \sigma_0 = 1/5$. Set $t = 1/250000$, and use the same kernel at each of the three groups:

$$w = (1/2, 1/2), \qquad L = \begin{pmatrix} 5/8 & 3/8 \\ 3/8 & 5/8 \end{pmatrix},$$

$$w' = (1/2 + t, 1/2 - t), \quad L' = \begin{pmatrix} 5/8 - t/4 & 3/8 - t/4 \\ 3/8 + t/4 & 5/8 + t/4 \end{pmatrix}.$$

These define $\theta, \eta \in \mathcal{K}$. All entries are positive and normalized; L has singular values 1 and $1/4$, and $\|L' - L\|_{\mathrm{op}} = t/2$ gives $\sigma_{\min}(L') \ge 1/4 - t/2 > 1/5$.

The two possible full-array alignments have squared distances

$$\|\theta - \eta\|_2^2 = \frac{11t^2}{4}, \qquad \|\theta - S_s\eta\|_2^2 = \frac{3}{4} + \frac{11t^2}{4}.$$

At these margins,

$$B^2 = 11171875 + 7593750\sqrt{2}, \qquad \rho^2 = \frac{1}{1152(11171875 + 7593750\sqrt{2})}.$$

Using $\sqrt{2} > 7/5$ gives $\rho^2 < 11t^2/4$, so $(\theta, \eta) \in Q_\rho$. Direct expansion of the eight observation cells gives

$$\|\mathcal{O}(\theta) - \mathcal{O}(\eta)\|_2^2 = \frac{t^2}{2048} + \frac{5t^4}{256} + \frac{t^6}{128} < (9 \times 10^{-8})^2. \tag{7.8}$$

For an alternative calculation, write the two reference columns as $c \pm (s/2)e$, with $c = (1/2, 1/2)^{\mathsf{T}}$, $e = (1, -1)^{\mathsf{T}}$, and $s = 1/4$. The perturbed columns are shifted by $-st\,e$. The one-way marginals agree; orthogonality of the tensor products of c and e reduces the squared difference to $6s^4t^4 + 8s^6(t/2 - 2t^3)^2$, which is (7.8).

Every valid uniform separation value must therefore satisfy $\Delta < 9 \times 10^{-8}$. For the particular sufficient rule in Corollary 7.3, $a \le \Delta/4$ and $m = 8$ imply

$$N \ge \frac{20480}{\Delta^2} > 2.528 \times 10^{18}.$$

If $\Delta = 2^{-k}$, then $k \ge 24$, and consequently

$$N \ge 20480\, 2^{48} = 5\,764\,607\,523\,034\,234\,880.$$

These are lower bounds on that prescribed sufficient budget per edge, not on the samples required by every estimator or by a sharper analysis of this model. In particular, they are not information-theoretic or minimax lower bounds.

Example 7.5 (A finite boundary model)**.** For the same binary one-edge design, now take $w_0 = 1/2$ and $\sigma_0 = 1$. Both weights must equal $1/2$. Each nonnegative normalized kernel column has squared norm at most one, whereas the Gram condition forces it to be at least one. It is therefore a coordinate vector. The two columns must be distinct, so every kernel is either I_2 or the swap matrix. Thus $\mathcal{K}$ has exactly eight arrays.

Index these arrays by $b \in \{0,1\}^3$, the three kernel choices. The observed law assigns mass $1/2$ to each of b and its bitwise complement $\bar{b}$. The four unordered support pairs $\{b, \bar{b}\}$ are disjoint, and are exactly the four W-orbits. Distinct orbits have quotient distance 2 and observation distance 1. Thus the analytic separation value $\Delta = 1$ is valid.

Here $D = \sqrt{14}$ and $B^2 = 157 + 108\sqrt{2}$. The rule in Corollary 7.3 becomes

$$N = \left\lceil \frac{40960}{7}(157 + 108\sqrt{2}) \right\rceil = 1\,812\,393.$$

Nevertheless, one observed triple determines its unique support pair, hence the true orbit, with probability one under this boundary model. This contrast illustrates conservatism of the general prescription. Its admissible set has zero seven-dimensional volume, so Proposition 6.5 supplies no nontrivial leaf count here. The analytic proof of $\Delta = 1$ also does not construct a serialized box certificate.

The metric d_W identifies every componentwise permutation, including permutations that exchange only part of two truth blocks. Therefore (7.5) does not itself bound a named truth sensitivity or prevalence. A functional transfer is valid when one has a separately justified inequality

$$|f(\theta) - f(\eta)| \le L_f d_\Gamma(\theta, \eta)$$

on the relevant set and a rate for that same gauge Γ. Such an inequality implies gauge invariance. Section 4 shows why it cannot be assumed for arbitrary truth functionals with $\Gamma = W$.

7.2. **Certifying semantic direction from gold samples.** Fix $J_0 \ge 1$ semantic test records before sampling. For record a, fix a group, event, sign $s_a \in \{-1, 1\}$, and positive sample sizes $n_{a,0}, n_{a,1}$. Let

$$p_{a,t} = \Pr(Y_{g_a} \in B_a \mid T = t), \qquad D_a = s_a(p_{a,1} - p_{a,0}).$$

The gold sample space is a product of two-stratum bit-string blocks. Its probability law is required, for each record a, to have the full block marginal

$$\Pr(Z_{a,0} = z_0, Z_{a,1} = z_1) = \prod_{t=0}^{1} \prod_{i=1}^{n_{a,t}} p_{a,t}^{z_{t,i}} (1 - p_{a,t})^{1 - z_{t,i}}. \tag{7.9}$$

Thus the two strata are independent within that block, and each stratum is IID. Dependence between records, and with observation blocks in a joint experiment, is unrestricted as long as the required block marginals hold. Metadata and sample sizes are fixed; the results do not invoke optional stopping or data-selected events.

Write $\widehat{p}_{a,t} = n_{a,t}^{-1} \sum_i Z_{a,t,i}$,

$$\widehat{D}_a = s_a(\widehat{p}_{a,1} - \widehat{p}_{a,0}), \qquad v_a = 1/n_{a,1} + 1/n_{a,0}.$$

A direction is certified at radius h only if $\widehat{D}_a - h > 0$; equality is abstention.

Proposition 7.6 (Gold direction and power bounds)**.** *Under* (7.9)*, for every* $u \ge 0$*,*

$$\Pr(\widehat{D}_a - D_a \ge u), \quad \Pr(\widehat{D}_a - D_a \le -u) \;\le\; \exp(-2u^2/v_a). \tag{7.10}$$

For $0 < \alpha < 1$*, let* $h_a(\alpha) = \sqrt{(v_a/2)\log(J_0/\alpha)}$*. Then*

$$\Pr\{\exists a : D_a < \widehat{D}_a - h_a(\alpha)\} \le \alpha, \tag{7.11}$$

$$\Pr\{\exists a : D_a \le 0,\ \widehat{D}_a - h_a(\alpha) > 0\} \le \alpha.$$

If $D_a \ge \gamma_a > 0$ *and* $\gamma_a \ge h_a(\alpha) + h_a(\beta)$ *for all* a*, with* $0 < \beta < 1$*, then*

$$\Pr\{\exists a : \widehat{D}_a - h_a(\alpha) \le 0\} \le \beta. \tag{7.12}$$

Under $D_a \geq \gamma > 0$ for every a, with equal stratum sizes n, sufficient integer sample sizes for the raw empirical-sign bound $\Pr\{\exists a : \widehat{D}_a \leq 0\} \leq \alpha$ and for successful certification (7.12)*, respectively, are*

$$n \geq \max\left\{1, \left\lceil \frac{\log(J_0/\alpha)}{\gamma^2} \right\rceil \right\}, \tag{7.13}$$

$$n \geq \max\left\{1, \left\lceil \frac{(\sqrt{\log(J_0/\alpha)} + \sqrt{\log(J_0/\beta)})^2}{\gamma^2} \right\rceil \right\}. \tag{7.14}$$

Proof. The centered contrast is a weighted sum of independent centered Bernoulli variables in one block. Their weights have squared sum v_a. The Bernoulli moment-generating bound, proved in Appendix C, gives (7.10). A union bound at $u = h_a(\alpha)$ gives (7.11). False positive certification is contained in that event. If certification fails under the stated margin, then

$$\widehat{D}_a - D_a \leq h_a(\alpha) - D_a \leq -h_a(\beta).$$

The non-strict lower-tail bound and a union bound give (7.12), including equality in the planning threshold. Separately, $\Pr(\widehat{D}_a \leq 0) \leq \exp(-2\gamma_a^2/v_a)$. When both stratum sizes are n, $v_a = 2/n$, giving (7.13); solving the power-margin inequality gives (7.14). □

The same finite law gives the exact count-pair distribution

$$\Pr(K_{a,0} = k_0, K_{a,1} = k_1) = \binom{n_{a,0}}{k_0} p_{a,0}^{k_0}(1-p_{a,0})^{n_{a,0}-k_0} \binom{n_{a,1}}{k_1} p_{a,1}^{k_1}(1-p_{a,1})^{n_{a,1}-k_1}.$$

This follows by counting bit strings and remains valid at $p = 0, 1$ with the usual convention $x^0 = 1$. Sample sizes in (7.13)–(7.14) count draws per record per truth stratum. Counting unique physical cases requires the sampling design. For binary rank-two truth models, component coverage in Corollary 3.5 and statistical direction certification can be used together. For larger coarsenings, certified directions enter the surviving-permutation test; they do not replace it.

7.3. **Confidence sets under approximate conditional independence.** Return to the binary truth model Θ in Corollary 3.5, with rank-two kernels K_g. Fix $\theta_* \in \Theta$. For each edge and truth value, let $Q_{e,t}$ be a normalized joint law whose single-group marginals are the target kernels $K_g(\cdot, t)$. Define

$$R_{e,t}(x) = \prod_{g \in e} K_g(x_g, t), \qquad P_e = \sum_t \pi_{[e]}(t) Q_{e,t}.$$

Supply $\varepsilon_{e,t} \in [0,1]$. Suppose $\mathrm{TV}(Q_{e,t}, R_{e,t}) \leq \varepsilon_{e,t}$ and supply deterministic $b_e \in [0,1]$ satisfying

$$\sum_t \pi_{[e]}(t) \varepsilon_{e,t} \leq b_e. \tag{7.15}$$

For example, $\max_t \varepsilon_{e,t}$ is a valid bound. The condition keeps target truth-conditional marginal performance fixed while allowing within-edge dependence beyond the product reference.

Use observation samples from the actual P_e and gold samples with probabilities from those target marginals. Let $0 < \delta_O, \delta_G < 1$, $\delta_O + \delta_G < 1$, and put

$$r_e = \sqrt{\frac{\log(2m/\delta_O)}{2N_e}}, \qquad \rho_e = \min\{1, d_e r_e/2\},$$

$$h_a^G = \sqrt{\frac{v_a}{2}\log(2J_0/\delta_G)}.$$

For each outcome, define the full-real set

$$\begin{aligned}\mathcal{S} = \{\vartheta \in \Theta :\ &\mathrm{TV}(\mathcal{O}(\vartheta)_e, \widehat{P}_e) \le \min\{1, b_e + \rho_e\} \quad \text{for all } e,\\ &|D_a(\vartheta) - \widehat{D}_a| \le h_a^G \quad \text{for all } a\}.\end{aligned} \tag{7.16}$$

An external deterministic anchor may also be intersected with this set if its truth at θ_* is separately supplied. The gold constraints themselves do not require a positive true direction.

Proposition 7.7 (Set and functional coverage)**.** *Under these assumptions,*

$$\mathrm{TV}(P_e, \mathcal{O}(\theta_*)_e) \le \sum_t \pi_{[e]}(t)\varepsilon_{e,t} \le b_e, \tag{7.17}$$

and $\Pr(\theta_* \in \mathcal{S}) \ge 1-\delta_O-\delta_G$*. On this event, for every probability-valued functional* $f : \Theta \to [0,1]$*, simultaneously,*

$$f(\theta_*) \in \left[\inf_{\vartheta\in\mathcal{S}} f(\vartheta),\ \sup_{\vartheta\in\mathcal{S}} f(\vartheta)\right]. \tag{7.18}$$

For any $\vartheta, \vartheta' \in \mathcal{S}$*,*

$$\mathrm{TV}(\mathcal{O}(\vartheta)_e, \mathcal{O}(\vartheta')_e) \le \min\{1, 2(b_e + \rho_e)\}. \tag{7.19}$$

Proof. Convexity of the ℓ_1 norm gives (7.17). Cellwise Hoeffding bounds and a union bound over all m cells give, with probability at least $1 - \delta_O$, $\mathrm{TV}(\widehat{P}_e, P_e) \le \rho_e$ on every edge. The two tails in (7.10) and a union bound give $|\widehat{D}_a - D_a(\theta_*)| \le h_a^G$ for every record with probability at least $1 - \delta_G$. On their intersection, triangle inequalities put θ_* in (7.16). This one parameter-in-set event implies all statements in (7.18), without another multiplicity penalty. The triangle inequality through $\widehat{P}_e$, together with the universal TV bound one, gives (7.19). □

The functional statement is pointwise on the single event $\theta_* \in \mathcal{S}$. Consequently it also permits a readout f chosen after seeing the data: the finite sample space presents no additional measurability issue. This does not permit data-dependent changes to the anchors, events, strata, sample sizes, or stopping rule used to construct $\mathcal{S}$; their coverage still depends on the fixed sampling assumptions.

The set Θ is not assumed compact, so the extrema in (7.18) are infima and suprema. A computed outer interval is valid if it contains $f(\vartheta)$ for every real $\vartheta \in \mathcal{S}$. A finite-grid range alone has no such coverage property. Equation (7.19) bounds observations; latent inversion additionally requires the explicitly conditioned setting of Section 5 and the appropriate gauge.

8. Interpretation and further directions

8.1. **What the observation design determines.** A group output can encode several dependent judgments, so no factorization within a group is needed. The product assumption is across groups conditional on the stated latent variable. The original binary truth model and the subclass model express different substantive hypotheses: the former takes T itself as the product class, whereas the latter takes Z as the product class and defines T by coarsening. Corollary 3.5 and Theorem 4.2 explain why their semantic conclusions differ.

Under the rank hypotheses, each observed triple determines a local unordered decomposition. Overlaps transmit a common indexing of distinguishable columns along the edge-intersection graph. Distinct components retain independent gauges because the model supplies no cross-component joint observations or linking constraints. Theorem 3.2 records this information exactly at the level of fibers, and Theorem 4.1 then treats external meaning through the remaining permutations.

These results suggest two separate design questions. Adding overlapping unlabeled triples can remove relative subclass alignments. Adding semantic information can remove scientifically unacceptable regroupings. The full-joint obstruction in Theorem 4.2 demonstrates a case in which the latter is essential even after all output information is supplied. The finite image in Theorem 4.4 quantifies the resulting performance uncertainty rather than forcing a selected grouping to be treated as identified.

8.2. **Mathematical and computational guarantees.** The exact fiber statements describe what the observation model determines, and the inverse statements give sufficient quantitative guarantees on a conditioned class. The certificate theorem adds a terminating rational method for obtaining the separation constant. These are different conclusions: uniqueness does not supply a practical estimator, a sound checker does not by itself establish successful search, and eventual search success does not bound its physical resource use.

The accompanying Lean development covers the principal identification, semantic, conditioning, and statistical interfaces, together with the exact compiler, certificate checker, and terminating search. Appendix D identifies their formal counterparts and separates the additional conventional proofs in this article. The checks concern the encoded statements under their hypotheses; scientific validity of the model remains an application-specific question.

8.3. **Limitations and future work.** The exact comparator count r, positivity of mixture weights, shared group kernels, and componentwise shared weights are central parts of the model. Extending the fiber description to unknown or overfitted state counts requires an analysis of splitting and merging of product components. A characterization for general rank-degenerate triples would also require local alternatives beyond the local-rigidity premise of Theorem 3.2. General coupled-decomposition criteria may offer stronger designs than imposing Kruskal's inequality on every edge.

The diagonal family yields an exact finite semantic image because its full fiber collapses to bijections. For general observations, a useful next objective is certified sharp sensitivity and specificity images under scientifically justified semantic constraints. Real-algebraic elimination provides a general framework for encoded polynomial problems [2]; an effective, verified sharp-image implementation is a stronger target than the positive-separation algorithm proved here. Related questions include finding which additional calibrated events or atom-membership facts eliminate all nonsemantic permutations, and optimizing their acquisition cost.

The explicit inverse constants use full-column singular-value margins. Extending comparable constants to robust Kruskal conditions, improving their dependence on dimensions and overlap structure, and exploiting design-specific redundancy could sharpen the quantitative results. For the present product-box certificate, Proposition 6.5 proves an output-size obstruction that improvements in traversal or pruning cannot eliminate. Different certificate geometries, quotient-domain rules, or compact proof representations merit investigation; their practicality is not established here. Unary serialization imposes an additional, independent cost. Example 7.4 quantifies the conservatism of the stated sufficient sample prescription, while Example 7.5 shows that an analytic model-specific analysis can be far sharper. The formal existence of the lexicographic estimator does not supply that fitting implementation.

Finite-sample guarantees assume fixed within-block IID laws, prespecified metadata, and deterministic discrepancy and conditioning bounds. Adaptive assignments, selected anchors, informative missingness, or estimated margins require additional probability arguments. Application-specific evaluation should separately assess observable fit, subclass recovery, semantic uncertainty, and computational cost. Resource-limited full-model prototype searches have not produced

a positive box certificate. Practical full-model certification, generally sharp semantic-image computation, numerical estimation studies, and real-data validation remain open tasks.

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Appendix A. An elementary full-column-rank uniqueness proof

For completeness, the full-column-rank input to the conditioned theory can be established without the general Kruskal theorem.

Proposition A.1. *Let*

$$P = \sum_{i=1}^{r} w_i a_i \otimes b_i \otimes c_i$$

have positive weights and stochastic factor columns, and suppose $A = (a_i)$, $B = (b_i)$, and $C = (c_i)$ all have column rank r. Any other positive normalized r-term stochastic representation of P agrees after a common permutation of its weights and factor columns.

Proof. Let A', B', C', w' be any such competitor. Summing over the third outcome gives

$$A \operatorname{diag}(w) B^{\mathsf{T}} = A' \operatorname{diag}(w') B'^{\mathsf{T}}.$$

The left side has rank r, because left inverses of A and B reduce it to the invertible matrix $\operatorname{diag}(w)$. Hence both A' and B' have rank r. The other two pairwise marginals similarly give rank r for C'. The displayed matrix has the same column space as A and A', and its transpose has the same column space as B and B'. Therefore

$$A' = AU, \qquad B' = BV$$

for invertible $r \times r$ matrices U, V.

Contract the first two tensor modes by left inverses of A, B. The common tensor becomes

$$\sum_i w_i e_i \otimes e_i \otimes c_i = \sum_j w'_j u_j \otimes v_j \otimes c'_j,$$

where u_j, v_j are the columns of U, V. Since C' has full column rank, for each j there is a linear functional taking c'_j to one and every other $c'_{j'}$ to zero. Contracting the third mode by this functional shows that $w'_j u_j v_j^{\mathsf{T}}$ is a nonzero diagonal rank-one matrix. Such a matrix has exactly one nonzero diagonal coordinate: if u_i and v_k are nonzero, diagonality forces $i = k$, and this forces both supports to be that singleton. Thus

$$u_j = \alpha_j e_{\pi(j)}, \qquad v_j = \beta_j e_{\pi(j)}$$

with nonzero scalars. Invertibility makes π a permutation. Column normalization of $A' = AU$ and $B' = BV$ gives $\alpha_j = \beta_j = 1$.

Comparing the (i, i) slice yields $w'_j c'_j = w_i c_i$ for the unique j with $\pi(j) = i$. Summing entries gives $w'_j = w_i$, and positivity then gives $c'_j = c_i$. This is the required common permutation. Notice that the competitor ranks were derived from the observations. □

Appendix B. Scalar representation and interval convergence

B.1. **Adequacy of derivative-certified brackets.** Let α be real algebraic over $\mathbb{Q}$. Choose a nonzero polynomial of minimal degree over $\mathbb{Q}$ vanishing at α. Its derivative cannot also vanish at α: otherwise their greatest common divisor would give a polynomial of smaller degree, since characteristic zero excludes an identically zero derivative. Clear denominators to obtain an integer polynomial p with $p(\alpha) = 0$ and $p'(\alpha) \neq 0$.

Continuity gives a neighborhood of α where p' has one strict sign. Choose rational $l < \alpha < u$ in that neighborhood. Strict monotonicity gives opposite endpoint signs. Natural interval Horner evaluation of a fixed polynomial converges to its point value as the input interval shrinks; this follows by induction through additions and multiplications. Shrink the rational bracket further until the interval evaluation of p' excludes zero. This proves that every real algebraic number admits the code used in Section 6. Canonical finite coefficient lists remove irrelevant leading zero coefficients from the representation.

B.2. **Exact refinement and comparison.** Given a valid bracket, evaluate p exactly at its rational midpoint. If the midpoint is a root, switch to its rational code. Otherwise retain the half whose endpoint signs differ. The derivative certificate remains valid on a subinterval, and its unique root is retained. Each nonterminal refinement halves the width, so any requested positive rational precision is achieved in finitely many steps.

To compare the represented root to a rational t, first use its position relative to the bracket. When t is inside the bracket, compute $p(t)$ exactly. If it is zero, equality holds; otherwise $t \neq \alpha$ and eventual refinement separates them. Endpoint equality with a bracket endpoint is impossible because the endpoint polynomial values are nonzero. Hence strict positivity and weak upper-margin comparisons are decidable, including equality at $1/r$ or 1. The procedure never needs to decide whether a represented root is irrational. A rational root need not occur at a bisection midpoint, but comparison with that same rational still detects it directly.

B.3. **Outward square roots and literal constants.** For a rational interval $[l, u]$ with $0 \leq l \leq u$ and rational dyadic mesh $h > 0$, choose grid points a, b by exact integer comparisons so that

$$a \leq \sqrt{l} < a + h, \qquad b - h < \sqrt{u} \leq b.$$

The interval $[a, b]$ encloses every square root from $[l, u]$ and has width at most $\sqrt{u - l} + 2h$. The construction uses integer square root after clearing the rational denominators. It includes zero and exact-square endpoints.

After positive preparation, the margin brackets have positive lower endpoints. Refine every current Root code to the specified stage width, changing it to a Rat code if an exact midpoint root is found. Use singleton intervals for Rat codes. Write $h_s = 2^{-s-2}$. The prepared interval for σ_0 contains $\sigma_0 \leq 1$ and has width at most $h_s \leq 1/4$, so its upper endpoint is at most $5/4$. Consequently the interval for σ_0^{-3}, and hence each β_e, has lower endpoint at least $64/125 > 1/2$. The radicand for each edge contribution to B has lower endpoint at least $(64/125)^2 > 1/4$. Since $1/2$ lies on every h_s mesh, the downward square-root endpoint is at least $1/2$, so the interval for B has lower endpoint at least $1/2$. Similarly $m \geq 1$ and the mesh contains 1, giving a lower endpoint at least 12 for M. Together with the positive margin intervals, these bounds make every reciprocal in the literal formulas legal at every stage, including $s = 0$. The intervals for κ and $\rho = \min\{1, \kappa/M\}$ also have positive lower endpoints.

The refined margin intervals remain in compact positive neighborhoods of their exact values, where reciprocals are bounded and continuous. All operations defining $a_g, \beta_e, B, \kappa, M, \rho$ preserve enclosure; square roots, minima and maxima are continuous. Induction through these literal

formulas gives coefficient intervals containing their exact values with widths tending to zero. Polynomial interval evaluation then gives Lemma 6.3 uniformly on bounded coordinate boxes. The argument allows dependence between repeated occurrences of a scalar, because enclosure uses a superset of all consistent assignments.

B.4. **Why all principal minors are required.** For a real symmetric matrix A, positive semidefiniteness implies nonnegative determinants of all principal submatrices. Conversely, suppose all principal minors are nonnegative. For $t > 0$, each principal determinant of $A + tI$ is a sum of principal minors of A times powers of t, including the strictly positive highest power. Its leading principal minors are therefore positive, so Sylvester's positive-definiteness criterion gives $A + tI \succ 0$. Letting $t \downarrow 0$ yields $A \succeq 0$. This proves the criterion used by the compiler on singular boundaries as well as in the interior. Testing only the leading minors of A would not suffice for semidefiniteness.

Appendix C. Finite probability bounds

For a Bernoulli variable X with mean p, define

$$\psi(\lambda) = \log \mathbb{E} e^{\lambda(X-p)}.$$

For $0 < p < 1$, differentiating twice identifies $\psi''(\lambda)$ with the variance of a Bernoulli variable under exponential tilting. It is at most $1/4$. Since $\psi(0) = \psi'(0) = 0$, integration gives $\psi(\lambda) \le \lambda^2/8$ for every real λ. The cases $p = 0, 1$ are immediate. Thus, for independent Bernoulli variables X_i and real weights a_i,

$$\mathbb{E} \exp\left(\lambda \sum_i a_i (X_i - \mathbb{E}X_i)\right) \le \exp\left(\frac{\lambda^2}{8} \sum_i a_i^2\right).$$

Put $V_a = \sum_i a_i^2$. If $V_a = 0$, every weight is zero and the centered sum is identically zero; either closed tail has probability one at $u = 0$ and zero for $u > 0$. If $V_a > 0$ and $u > 0$, Markov's inequality optimized at $\lambda = 4u/V_a$ gives the upper-tail bound $\exp(-2u^2/V_a)$. For $u = 0$ the same bound is one and follows directly from the probability bound. Negating the weights gives the lower tail. The observation and gold applications have $V_a = 1/N_e > 0$ and $V_a = v_a > 0$, respectively. This is the Bernoulli specialization of Hoeffding's inequality [9].

For one empirical observation cell, $a_i = 1/N_e$, giving the exponent $-2N_e u^2$. The two tails at $u = t_e(\delta)$ have sum at most δ/m. A union bound over every edge cell proves (7.2). Zero-count cells remain included in m and in the event. For a gold contrast the weights are $s_a/n_{a,1}$ in truth stratum one and $-s_a/n_{a,0}$ in stratum zero, with squared sum v_a, giving (7.10). The full marginal laws in Section 7 justify the independence used within each of these calculations. Their union bounds hold under arbitrary coupling of different blocks.

Appendix D. Formal theorem correspondence

The following declarations identify the formal counterparts of the principal results. The companion documentation gives a detailed source map and build instructions.

Observational fibers: Theorem 3.2 corresponds to `Gen.fiber_eq_compatible_image` and `Gen.identifies_iff_compatible_mod_gauge`. Corollary 3.5 is the paper-level binary specialization of these results, with the strict-anchor argument in its proof. The general normalized Kruskal proof is `Gen.Kruskal.stochastic_uniqueness`; `Gen.KruskalConsequences.GEN_G_KRUSKAL` supplies the design specialization of Theorem 3.4.

Semantic fibers: Theorem 4.1 is represented by `Gen.KruskalConsequences.GEN_G_SEMANTIC`, and Theorem 4.2 by `Gen.NoSemantic.GEN_G_NOSEM`. The all-real diagonal fiber and its images appear in `Gen.Diagonal.full_diagonal_fiber`, `Gen.Diagonal.full_anchored_images`, and `Gen.Diagonal.full_anchored_sharp`. The four-state full-fiber specialization is `Ext.Hidden.GEN_F_STAR`.

Conditioning: The explicit derivative bounds are `Gen.Jacobian.observation_jacobian_lower_bound` and `Gen.Hessian.observation_hessian_bound`. The literal local and global inverse interfaces and rate are in `Gen.ConditionedRates.literal_local_inverse`, `Gen.ConditionedRates.literal_global_inverse`, and `Gen.ConditionedRates.literal_rate_stat`. The formal lexicographic baseline establishes existence and all-outcome optimality. Example 7.5 uses the existing finite-control results `Gen.FiniteControl.full_real_K_has_eight` and `Gen.FiniteControl.prescribed_count_exact`.

Positive separation: The full-real compiler proves `Cover.ConcreteCompiler.coverEncoding_iff` and `Cover.FullRealBridge.concrete_coverEncoding_iff_fullDistance`. The instantiated checker theorem is `Cover.A06.verifyCoverCertificate_sound`. The stage theorem is `Cover.A06.coverStage_eventually_succeeds`. The byte-to-real chain is `Cover.A06.runCover_success_fully_checked`, `Cover.A06.verifyCertificateBytes_sound`, then `Cover.A06.verifyCoverCertificate_sound`: exact input/output binding yields byte-checker acceptance, which yields the full-real separation inequality. For the finite-fuel program, `Cover.A06.runCoverCapped_success_separates` gives the real separation conclusion directly. The downstream inverse and statistical interfaces include `Cover.A06.conditioned_global_inverse_of_cover` and `Cover.A06.conditioned_rate_stat_of_cover`.

Checker soundness applies to arbitrary accepted certificates, independently of the generator and without assuming a positive gap. Eventual search success additionally uses the compiled model, compactness, and conditioned uniqueness.

The finite-fuel interface is total, prefix stable, and eventually successful. These claims concern exact rational computation and complete certificate replay, not physical resource bounds.

Scope of the conventional arguments. Proposition 6.5, the volume and rational-witness calculations in Examples 6.6 and 7.4, the illustrative Examples 3.3 and 4.6, and the bounded-differences refinement in Remark 7.2 are proved in this article and are not asserted to have Lean counterparts in the accompanying source. The arbitrary, possibly data-selected functional readout following Proposition 7.7 is a pointwise consequence of set inclusion; the existing formal statistical API states a finite-family interface. No additional formal coverage is implied by these extensions.

Reproducible formal scope. The source fixes Lean 4.32.0 and mathlib revision `81a5d257c8e410db227a6665ed08f64fea08e997`. The permitted logical axioms are propositional extensionality, classical choice, and quotient soundness. The separate reproducibility documentation records dependency handling and supplementary numerical checks.

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